Examiner only

6. *(a)* Fertilisers usually contain compounds of three essential elements required for healthy and productive plant growth. The table shows the properties of two nitrogenous fertilisers.

Property	Ammonium nitrate	Urea
% nitrogen	34	46
Mass of fertiliser absorbed by plants per 2000 litre spreader (kg)	690	560
Solubility	very soluble	very soluble
Cost (£/tonne)	360	350
Loss of ammonia due to evaporation	low	high
Weather dependency	effective in all conditions	effectiveness dependent on specific temperature and rainfall conditions

(i) Which fertiliser leads to the absorption of more **nitrogen** from a full 2000 litre spreader? Show your working. [2]

Fertiliser ....

(ii) Urea is the most widely used fertiliser in the world. However, many British farmers prefer to use ammonium nitrate.

Put a tick (✓) in the box next to the statement which suggests the **main** reason for this. [1]

- ammonium nitrate contains less nitrogen than urea ☐
- ammonium nitrate is better suited to British weather conditions than urea ☐
- more ammonium nitrate is absorbed by plants than urea ☐
- ammonium nitrate is more expensive than urea ☐

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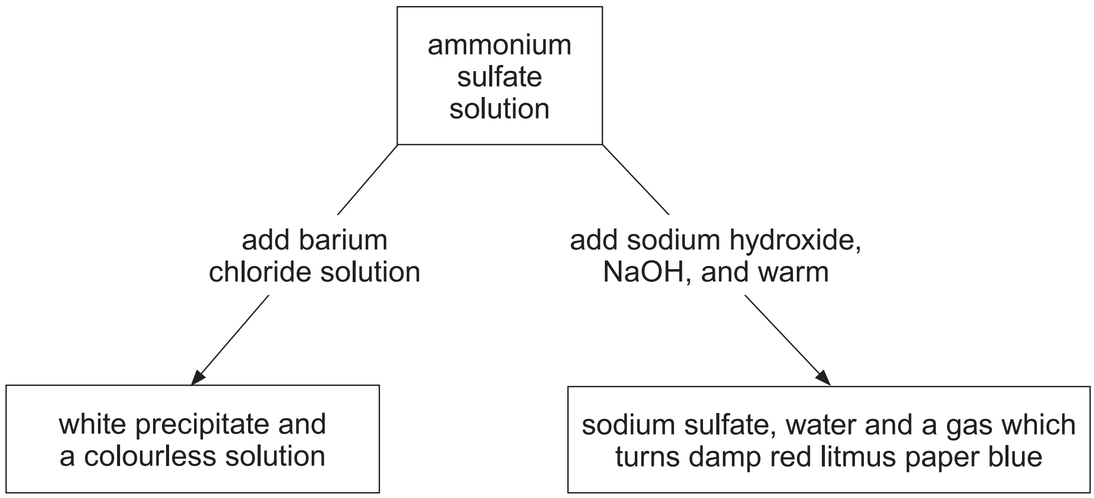
(b) The flow chart shows the chemical tests for the identification of the ions present in ammonium sulfate, $(NH_4)_2SO_4$.



(i) Write a balanced **symbol** equation for the reaction between ammonium sulfate and sodium hydroxide. [2]

....

(ii) The symbol equation below represents the reaction occurring between ammonium sulfate solution and barium chloride solution.

$$BaCl_2(aq) + (NH_4)_2SO_4(aq) \longrightarrow 2NH_4Cl(aq) + BaSO_4(s)$$

Write the **ionic** equation for the reaction. Include the state symbols. [2]

.... + ... ⟶ ....

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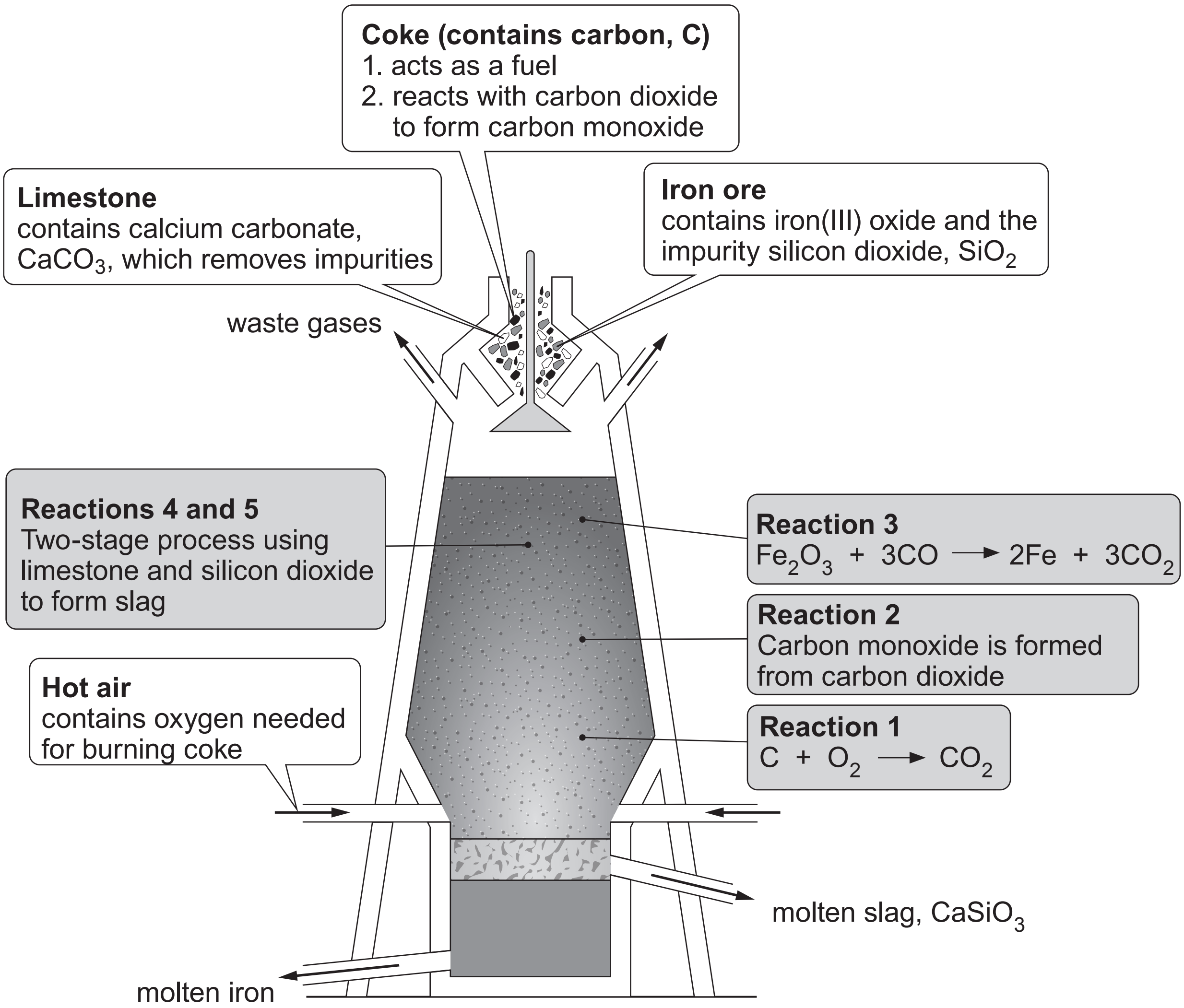
6. *(a)* Iron is extracted from its ore in the blast furnace.



Use information from the diagram and your knowledge to answer parts (i) and (ii).

(i) Write a balanced symbol equation for reaction **2**. [2]

....

(ii) Describe the two-stage process to form slag. [3]

....

....

....

....

Examiner only

(b) Iron(III) oxide reacts with dilute hydrochloric acid forming iron(III) chloride and water.

(i) Balance the symbol equation for this reaction. [1]

$$Fe_2O_3(s) + \boxed{}\,HCl(aq) \longrightarrow \boxed{}\,FeCl_3(aq) + \boxed{}\,H_2O(l)$$

(ii) Sodium hydroxide solution can be used to detect the presence of aqueous iron(III) ions.

The symbol equation below represents the reaction occurring between solutions of sodium hydroxide and iron(III) chloride.

$$3NaOH(aq) + FeCl_3(aq) \longrightarrow 3NaCl(aq) + Fe(OH)_3(s)$$

Write the **ionic** equation for the formation of the precipitate. [2]

....

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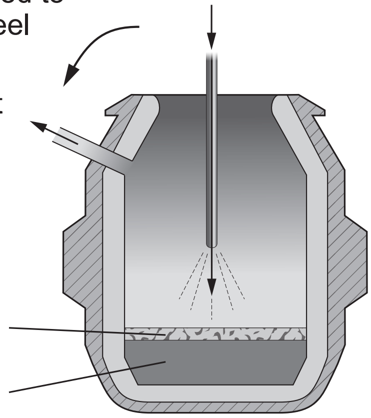
(c) **Steel manufacture in the UK**



Basic Oxygen Furnace

One of the major steelmaking processes used today in the UK is the Basic Oxygen Furnace, BOF. The raw materials for the BOF are cast iron from a blast furnace and scrap steel. Oxygen (>99.5 % pure) is "blown" into the BOF at supersonic speed. The impurities are oxidised producing great quantities of heat which melts the scrap steel.

Steel can be described in general terms as iron containing small amounts of carbon, to make it tougher and more ductile. There are many types of steel, each with its own specific chemical composition and properties to meet the needs of the many different applications. **Figure 1** below shows the relationships between the carbon content of steel and its ductility, tensile strength and hardness.

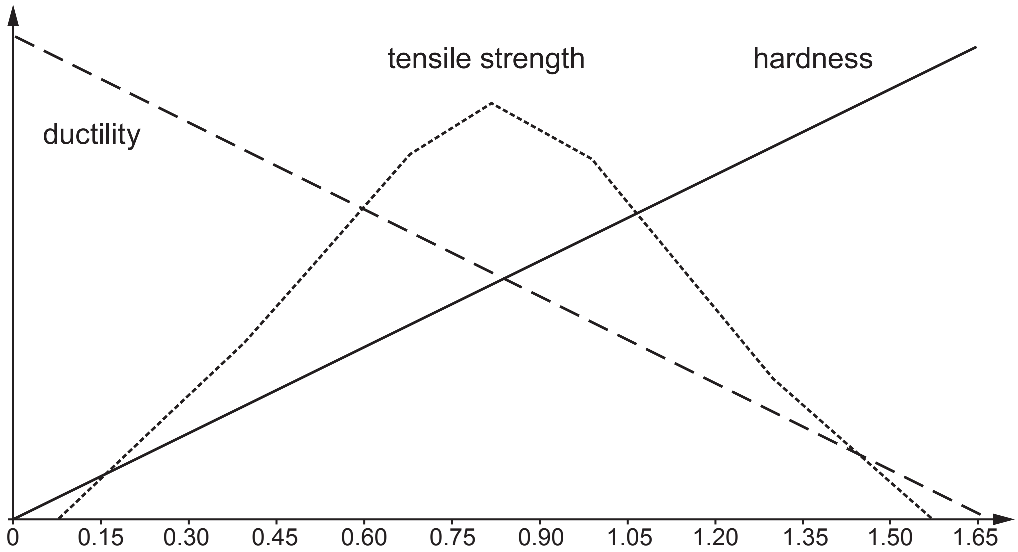


Figure 1

Ductility is a material's ability to be pulled into a wire.

Tensile strength is a measurement of the force required to pull something such as wire to the point where it breaks.

Hardness is a measure of how resistant a material is to permanent shape change when a compressive force is applied.

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Figure 2 shows the percentage of carbon in various alloys of iron.

Name of alloy	Percentage of carbon
low carbon steel	0.0 – 0.6 %
medium carbon steel	0.6 – 0.8 %
high carbon steel	0.8 – 1.3 %
very high carbon steel	1.3 – 1.6 %

Figure 2

(i) Tick (✓) the box next to the statement which best describes **one** way that production costs are reduced. [1]

high purity oxygen is used ☐

impurities are oxidised forming heat ☐

oxygen is blasted in at supersonic speed ☐

scrap steel is used in the process ☐

(ii) Tick (✓) the box next to the statement which best describes the effect of increasing the percentage of carbon in steel from 0 % to 0.45 %. [1]

ductility increases, hardness increases ☐

tensile strength increases, ductility increases ☐

ductility decreases, tensile strength increases ☐

hardness increases, tensile strength decreases ☐

(iii) A steel manufacturer wants to design an alloy with a high tensile strength but low ductility. Tick (✓) the box next to the approximate value for the percentage of carbon that should be included. [1]

0.2 ☐ 0.6 ☐ 1.0 ☐ 1.5 ☐

(iv) Name the alloy which is the most easily pulled into a wire and withstands the least compressive force. [1]

....

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Examiner only

6. (a) Kerosene is one of the fractions separated during the fractional distillation of crude oil.

Kerosene contains dodecane, $C_{12}H_{26}$. Dodecane undergoes a further process **A** to form smaller, more useful hydrocarbons, ethene, butene and one molecule of hydrocarbon **X**.

(i) Name process **A**. [1]

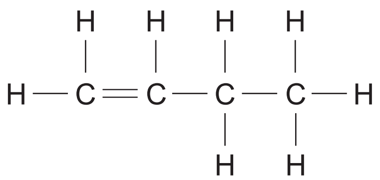
....

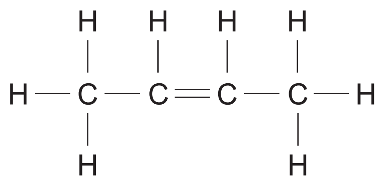
(ii) Complete the equation for this reaction. [1]

$$C_{12}H_{26} \longrightarrow C_2H_4 + C_4H_8 + \ldots\ldots\ldots\ldots$$

dodecane ethene butene hydrocarbon **X**

(b) C_4H_8 has three isomers. The diagrams below show two of the three isomers.

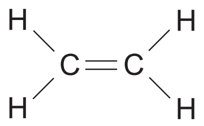




The third isomer is methylpropene. Draw its structure. [1]

Examiner only

(c) (i) Polyethene is made from ethene by addition polymerisation. Draw the structure of the repeating unit for polyethene. [1]

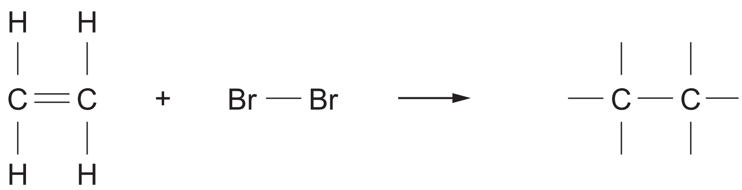


ethene

repeating unit for polyethene

(ii) The presence of the double bond in ethene can be confirmed using bromine water.

I. Complete the equation for the reaction between ethene and bromine. [1]



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II. Tick (✓) the box next to the name of the product formed. [1]

1,2-dibromoethene ☐

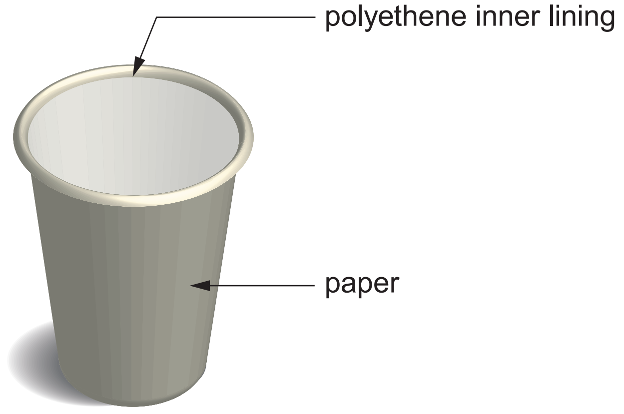
1,1-dibromoethane ☐

1,2-dibromoethane ☐

1,1-dibromoethene ☐

Examiner only

(d) In the UK, around 3 billion disposable coffee cups are used every year. Each paper coffee cup is lined with polyethene which is impossible to remove at a recycling plant. If the automated machine that sorts waste at a recycling centre detects a plastic lining, it rejects the cup sending it to general household waste.



(i) Most general household waste is disposed of in landfill sites.

Apart from using landfill sites, give the **other** disposal method of general household waste. State an environmental problem associated with the method. [2]

Method ....

Problem ....

....

(ii) Recycling helps to conserve raw materials. Name the raw material used to make polyethene. Give the main reason why it is important to conserve this raw material. [2]

Raw material ....

Main reason ....

....

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 (3410UB0-1)

Examiner only

6. (a) The table shows the formulae of the first four members of the alcohol family.

Alcohols
CH_3OH
C_2H_5OH
C_3H_7OH
C_4H_9OH

Give the general formula for the alcohol family. [1]

....

(b) The following word equation represents the formation of ethanol by fermentation.

$$\text{glucose solution} \xrightarrow{\text{yeast}} \text{ethanol solution} + \text{carbon dioxide gas}$$

(i) Complete and balance the equation for the production of ethanol and carbon dioxide from glucose.

Include the state symbols for the reactant and the products. [3]

$$C_6H_{12}O_6 \,(\ldots\ldots) \longrightarrow 2 \ldots\ldots\ldots\ldots \,(\ldots\ldots) + \boxed{}\, CO_2 \,(\ldots\ldots)$$

(ii) Yeast contains a substance which catalyses the breakdown of glucose.

Give the **general** name of this type of catalyst. [1]

....

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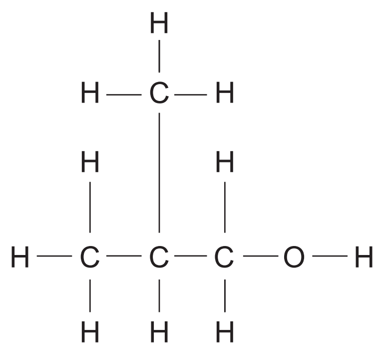
(c) (i) C_4H_9OH has four isomers. The diagram below shows the structure of one of the isomers.



Put a **tick** (✓) in the box next to the name of this isomer. [1]

butan-1-ol	☐
2-methylpropan-1-ol	☐
butan-2-ol	☐
2-methylpropan-2-ol	☐

(ii) Pentan-3-ol is one of the isomers of $C_5H_{11}OH$.

Draw the structure of pentan-3-ol. [1]

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 (3410UD0-1) **Turn over.**

Examiner only

6. (a) **X**, **Y** and **Z** are solutions of ethanoic acid, hydrochloric acid and sodium chloride, **but not necessarily in that order**. Reactions of **X**, **Y** and **Z** with magnesium gave the results shown below.

Solution	Reaction with magnesium
X	rapid fizzing, salt formed, temperature increase of 18 °C
Y	no reaction
Z	slow fizzing, salt formed, temperature increase of 11 °C

Use the table to identify **X**, **Y** and **Z**, giving your reasoning. Explain the results recorded and include equations to support your answer. [6 QER]

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(b) State how adding sodium hydroxide solution allows solutions containing iron(II) and iron(III) ions to be identified. Give the observations made for both solutions. [2]

....

....

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